



```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <string.h>

int main() {
    char command[100];

    printf("Enter a Linux command: ");
    scanf("%s", command);

    pid_t pid = fork();

    if (pid < 0) {
        printf("Fork failed!\n");
        return 1;
    }

    else if (pid == 0) {
        printf("Child Process\n");
        printf("Child PID : %d\n", getpid());

        printf("Executing command: %s\n", command);

        execlp(command, command, NULL);

        printf("Command execution failed!\n");
    }

    else {
        printf("Parent Process\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);

        wait(NULL);

        printf("Child process completed.\n");
    }

    return 0;
}
```



jhansi_priya@Jhansi-Priya: ~/



```
jhansi_priya@Jhansi-Priya:~$ mkdir OS_LAB
jhansi_priya@Jhansi-Priya:~$ cd OS_LAB
jhansi_priya@Jhansi-Priya:~/OS_LAB$ nano task1.c
jhansi_priya@Jhansi-Priya:~/OS_LAB$ gcc tas1.c -o task1
cc1: fatal error: tas1.c: No such file or directory
compilation terminated.
jhansi_priya@Jhansi-Priya:~/OS_LAB$ gcc task1.c -o task1
jhansi_priya@Jhansi-Priya:~/OS_LAB$ ./task1
Enter a Linux command: ls
Parent Process
Parent PID : 1222
Child PID  : 1245
Child Process
Child PID : 1245
Executing command: ls
task1 task1.c
Child process completed.
jhansi_priya@Jhansi-Priya:~/OS_LAB$
```

```
Child Process
Child PID : 1245
Executing command: ls
task1 task1.c
Child process completed.
jhansi_priya@Jhansi-Priya:~/OS_LAB$ uname -a
Linux Jhansi-Priya 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 x86_64 x86_64 GNU/Linux
jhansi_priya@Jhansi-Priya:~/OS_LAB$ lscpu
Architecture:                x86_64
CPU op-mode(s):              32-bit, 64-bit
Address sizes:               39 bits physical, 48 bits virtual
Byte Order:                  Little Endian
CPU(s):                      8
On-line CPU(s) list:        0-7
Vendor ID:                   GenuineIntel
Model name:                  11th Gen Intel(R) Core(TM) i7-1185G7 @ 3.00GHz
CPU family:                  6
Model:                      140
Thread(s) per core:         2
Core(s) per socket:         4
Socket(s):                   1
Stepping:                    1
BogoMIPS:                    3609.59
Flags:                       fpu vme de pse tsc msr pae mce cx8
                             apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtscp lm constant_tsc arch_perfmon rep_good nopl xt
```

se4_2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch ssbd ibrs ibpb stibp ibrs_enhanced tpr_shadow ept vpid ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid avx512f avx512dq rdseed adx smap avx512ifma clflushopt clwb avx512cd sha_ni avx512bw avx512vl xsaveopt xsavec xgetbv1 xsaves vnm1 avx512vbmi umip avx512_vbmi2 gfni vaes vpclmulqdq avx512_vnni avx512_bitalg avx512_vpopcntdq rdpid movdiri movdir64b fsrm avx512_vp2intersect md_clear ibt flush_l1d arch_capabilities

Virtualization features:

Virtualization:	VT-x
Hypervisor vendor:	Microsoft
Virtualization type:	full

Caches (sum of all):

L1d:	192 KiB (4 instances)
L1i:	128 KiB (4 instances)
L2:	5 MiB (4 instances)
L3:	12 MiB (1 instance)

NUMA:

NUMA node(s):	1
NUMA node0 CPU(s):	0-7

Vulnerabilities:

Gather data sampling:	Not affected
Ghostwrite:	Not affected

Tsx async abort: Not affected

Vmscape: Not affected

jhansi_priya@Jhansi-Priya:~/OS_LAB\$ ps

PID	TTY	TIME	CMD
281	pts/0	00:00:00	bash
2307	pts/0	00:00:00	ps

jhansi_priya@Jhansi-Priya:~/OS_LAB\$ ps -ef

UID	PID	PPID	C	STIME	TTY	TIME	CMD
root	1	0	0	14:07	?	00:00:02	/sbin/init
root	2	1	0	14:07	hvc0	00:00:00	/init
root	7	2	0	14:07	hvc0	00:00:00	plan9 --cont
root	53	1	0	14:07	?	00:00:00	/usr/lib/sys
root	98	1	0	14:07	?	00:00:00	/usr/lib/sys
systemd+	123	1	0	14:07	?	00:00:00	/usr/lib/sys
systemd+	131	1	0	14:07	?	00:00:00	/usr/lib/sys
root	165	1	0	14:07	?	00:00:00	/usr/sbin/cr
message+	166	1	0	14:07	?	00:00:00	@dbus-daemon
root	173	1	0	14:07	?	00:00:00	/usr/lib/sys
root	189	1	0	14:07	tty1	00:00:00	/sbin/agetty
syslog	194	1	0	14:07	?	00:00:00	/usr/sbin/rs
root	197	1	0	14:07	?	00:00:00	/usr/bin/pyt
root	279	2	0	14:07	?	00:00:00	/init
root	280	279	0	14:07	?	00:00:00	/init
jhansi_+	281	280	0	14:07	pts/0	00:00:00	-bash
root	282	2	0	14:07	pts/1	00:00:00	/bin/login -
jhansi_+	325	1	0	14:07	?	00:00:00	/usr/lib/sys
jhansi_+	326	325	0	14:07	?	00:00:00	(sd-pam)
jhansi_+	350	282	0	14:07	pts/1	00:00:00	-bash
root	2117	2	0	14:13	?	00:00:00	/init
root	2119	2117	0	14:13	?	00:00:00	/init
jhansi_+	2125	2119	0	14:13	pts/2	00:00:00	-bash

```

jhansi_priya@Jhansi-Priya:~/OS_LAB$ top
top - 14:17:58 up 15 min,  1 user,  load average: 0.02, 0.05, 0
Tasks:  24 total,   1 running,  23 sleeping,   0 stopped,   0 z
%Cpu(s):  0.1 us,   0.1 sy,   0.0 ni, 99.7 id,   0.0 wa,   0.0 hi,
MiB Mem :  7635.6 total,  7108.7 free,    500.1 used,    175.
MiB Swap:  2048.0 total,  2048.0 free,     0.0 used.   7135.

```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM
166	message+	20	0	9548	5460	4760	S	0.3	0.1
1	root	20	0	21732	13220	9860	S	0.0	0.2
2	root	20	0	3180	2204	2072	S	0.0	0.0
7	root	20	0	3196	2128	2008	S	0.0	0.0
53	root	19	-1	50308	17572	16488	S	0.0	0.2
98	root	20	0	25548	7000	5216	S	0.0	0.1
123	systemd+	20	0	21344	13248	11124	S	0.0	0.2
131	systemd+	20	0	91036	8052	7088	S	0.0	0.1
165	root	20	0	4244	2752	2496	S	0.0	0.0
173	root	20	0	17980	8892	7848	S	0.0	0.1
189	root	20	0	3124	2040	1900	S	0.0	0.0
194	syslog	20	0	222516	5780	4732	S	0.0	0.1
197	root	20	0	107032	23276	13820	S	0.0	0.3
279	root	20	0	3184	1108	980	S	0.0	0.0
280	root	20	0	3200	1248	1108	S	0.0	0.0
281	jhansi_+	20	0	6080	5428	3708	S	0.0	0.1
282	root	20	0	6704	4740	3952	S	0.0	0.1
325	jhansi_+	20	0	20324	11644	9516	S	0.0	0.1
326	jhansi_+	20	0	21164	3592	1868	S	0.0	0.0
350	jhansi_+	20	0	6080	5332	3668	S	0.0	0.1